Apunts Web

Copiar documento default.

```
cp /etc/apache2/sites-available/000-default /etc/apache2/sites-available/acme.conf
```

Borrar sites default

```
dmartinez@daniserver:/etc/apache2/sites-available$ sudo a2dissite 000-default.conf
```

Document /etc/apache2/sites-avaiable/(document web)

```
  GNU nano 7.2
<VirtualHost 192.168.1.20:80>
        ServerAdmin dmartinez@localhost
        DocumentRoot /var/www/IP180
        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>

<VirtualHost 192.168.1.21:80>
        ServerAdmin dmartinez@localhost
        DocumentRoot /var/www/IP280
        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>

<VirtualHost 192.168.1.21:8080>
        ServerAdmin dmartinez@localhost
        DocumentRoot /var/www/IP28080
        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>

<VirtualHost *:80>
        ServerAdmin dmartinez@localhost
        ServerName dani.local
        DocumentRoot /var/www/dani.local
        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>

<VirtualHost *:80>
        ServerAdmin dmartinez@localhost
        ServerName martinez.local
        DocumentRoot /var/www/martinez.local
        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

Habilitar pagina

```
dmartinez@daniserver:/etc/apache2/sites-available$ sudo a2ensite virtual.conf
```

Netplan

```
network:
    ethernets:
        enp0s3:
         addresses:
            - 192.168.1.20/24
            - 192.168.1.21/24
```

Crear carpetes

```
dmartinez@daniserver:/etc/apache2/sites-available$ cd /var/www/
dmartinez@daniserver:/var/www$ sudo mkdir IP180
dmartinez@daniserver:/var/www$ sudo mkdir IP280
dmartinez@daniserver:/var/www$ sudo mkdir IP8080
dmartinez@daniserver:/var/www$ sudo mkdir dani.local
dmartinez@daniserver:/var/www$ sudo mkdir martinez.local
dmartinez@daniserver:/var/www$ rmdir IP8080
rmdir: failed to remove 'IP8080': Permission denied
dmartinez@daniserver:/var/www$ sudo rmdir IP8080
dmartinez@daniserver:/var/www$ sudo mkdir IP28080
dmartinez@daniserver:/var/www$ _
```

Fitxer /etc/apache2/ports.conf

```
  GNU nano 7.2                                                /etc/apache2/ports.conf *
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 80
Listen 8080

<IfModule ssl_module>
        Listen 443
</IfModule>

<IfModule mod_gnutls.c>
        Listen 443
</IfModule>
```

/etc/hosts

```
  GNU nano 7.2                                                        /etc/hosts *
127.0.0.1 localhost
127.0.1.1 daniserver
127.0.0.1 dani.local martinez.local www.dani.local www.martinez.local
```

Index html

```
dmartinez@daniserver:/var/www$ sudo nano dani.local/index.html_
```

Curl

```
dmartinez@daniserver:/var/www$ curl http://192.168.1.21:8080_
```

SSL

Crear ssl

sudo openssl req -x509 -nodes -newkey rsa:2048 -keyout /etc/ssl/private/private.key -out /etc/ssl/certs/cert.pem -days 365

Activar ssl



Rutes de SSL

```
SSLCertificateFile      /etc/ssl/certs/cert.pem
SSLCertificateKeyFile   /etc/ssl/private/private.key
```

```
<VirtualHost *:443>
        ServerAdmin dmartinez@localhost
        ServerName dani.smx.local
        DocumentRoot /var/www/dani.smx.local
        ErrorLog ${APACHE_LOG_DIR}/error1.log
        CustomLog ${APACHE_LOG_DIR}/access1.log combined
        SSLEngine on
        SSLCertificateFile      /etc/ssl/certs/cert.pem
        SSLCertificateKeyFile   /etc/ssl/private/private.key
        <FilesMatch "\.(?:cgi|shtml|phtml|php)$">
                SSLOptions +StdEnvVars
        </FilesMatch>
        <Directory /usr/lib/cgi-bin>
                SSLOptions +StdEnvVars
        </Directory>
</VirtualHost>

<VirtualHost *:443>
        ServerAdmin dmartinez@localhost
        ServerName aula23.smx.local
        DocumentRoot /var/www/aula23.smx.local
        ErrorLog ${APACHE_LOG_DIR}/error2.log
        CustomLog ${APACHE_LOG_DIR}/access2.log combined
        SSLEngine on
        SSLCertificateFile      /etc/ssl/certs/cert.pem
        SSLCertificateKeyFile   /etc/ssl/private/private.key
        <FilesMatch "\.(?:cgi|shtml|phtml|php)$">
                SSLOptions +StdEnvVars
        </FilesMatch>
        <Directory /usr/lib/cgi-bin>
                SSLOptions +StdEnvVars
        </Directory>
</VirtualHost>
```

Redireccionar

```
<VirtualHost *:80>
        ServerName dani.smx.local
        Redirect permanent / https://dani.smx.local
</VirtualHost>
<VirtualHost *:80>
        ServerName aula23.smx.local
        Redirect permanent / https://aula23.smx.local
</VirtualHost>
```

```
dmartinez@daniserver:/etc/apache2/sites-available$ sudo a2ensite virtual.conf
```

Comprovacions

```
  GNU nano 7.2                                                    /etc/hosts *
127.0.0.1 localhost
127.0.1.1 daniserver
127.0.0.1 dani.smx.local aula23.smx.local www.dani.smx.local www.aula23.smx.local_
```

```
dmartinez@daniserver:/etc/apache2/sites-available$ ls ../sites-enabled/
000-default.conf  ssl.conf  virtual.conf
```

a2ensite per activar

Comandes per a comprovacions:

Curl normal
Curl -k ssl

Errors

Crear carpetes i afegir missatge d'error

```
 57  sudo mkdir -p dani.local/error
 58  cd dani.local/
 59  sudo nano index.html
 60  cd error/
 61  sudo nano error.html
 62  cd /etc/apache2/sites-available/
 63  sudo nano virtual.conf
 64  sudo a2ensite virtual.conf
 65  sudo service apache2 restat
 66  sudo service apache2 restart
 67  sudo nano /etc/hosts
 68  curl http://dani.local
 69  curl http://dani.local/jdjas
```

virtual.conf

```
ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined
ErrorDocument 500 "Error 500"
ErrorDocument 404 /error/error.html
```

```
64  sudo a2ensite virtual.conf
65  sudo service apache2 restat
66  sudo service apache2 restart
67  sudo nano /etc/hosts
```

AUTENTICACIÓ

Creació dels fitxers

Creació dels fitxers

- Bàsicament, es tracta de crear un fitxer d'usuaris amb passwords. al directori on volem crear el fitxer, per exemple a /usr/local/apache/passwd

```
$roger@roger-desktop:~$ sudo htpasswd -c secret usuari

[sudo] password for roger:
New password:
Re-type new password:
Adding password for user usuari

roger@roger-desktop:~$ cat secret
usuari:$apr1$Ovsvd5mg$3A7xY44qPFvINpjqFjyHN1
```

- Si volem crear més usuaris, pels següents usuaris NO posem el paràmetre -c a htpasswd.
- I de posar dintre el directori que volem protegir:

```
dmartinez@daniserver:/etc/apache2/sites-available$ sudo mkdir -p /usr/local/apache/passwd
dmartinez@daniserver:/etc/apache2/sites-available$ sudo htpasswd -c /usr/local/apache/passwd/secret dmartinez
New password:
Re-type new password:
Adding password for user dmartinez
```

Crear el document a /usr/local/apache/passwd/group

```
  GNU nano 7.2                                                    group *
contribuidors: contribuidor
```

Fitxer amb autenticació

```
<VirtualHost *:80>
        ServerName dani.local
        ServerAdmin dmartinez@localhost
        DocumentRoot /var/www/dani.local
        ErrorLog ${APACHE_LOG_DIR}/error1.log
        CustomLog ${APACHE_LOG_DIR}/access1.log combined
        <Directory /var/www/dani.local>
         Options -Indexes
         AuthType Basic
         AuthName "Zona restringida, requereix usuari i password"
         AuthBasicProvider file
         AuthUserFile /usr/local/apache/passwd/secret
         Require user dmartinez profe
        </Directory>
</VirtualHost>

<VirtualHost *:80>
        ServerName public.local
        ServerAdmin dmartinez@localhost
        DocumentRoot /var/www/public.local
        ErrorLog ${APACHE_LOG_DIR}/error2.log
        CustomLog ${APACHE_LOG_DIR}/access2.log combined
        <Directory /var/www/public.local>
         AuthType Basic
         AuthName "Zona restringida, requereix usuari i password"
         AuthBasicProvider file
         AuthUserFile /usr/local/apache/passwd/secret
         AuthGroupFile /usr/local/apache/passwd/group
         Require group contribuidors
        </Directory>
</VirtualHost>
```

```
rocil@rocil:/etc/apache2/sites-available$ sudo a2enmod authz_groupfile
Considering dependency authz_core for authz_groupfile:
Module authz_core already enabled
Enabling module authz_groupfile.
To activate the new configuration, you need to run:
  systemctl restart apache2
```

sudo service apache2 restart o reboot al acabar

Comprovació

```
  GNU nano 7.2                                /etc/hosts *
127.0.0.1 localhost
172.16.24.113 dani.local www.dani.local
```

```
dmartinez@daniserver:/etc/apache2/sites-available$ a2ensite virtual.conf
```

```
dmartinez@daniserver:/etc/apache2/sites-available$ sudo a2ensite virtual.conf
Enabling site virtual.
To activate the new configuration, you need to run:
  systemctl reload apache2
dmartinez@daniserver:/etc/apache2/sites-available$ systemctl reload apache2
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to reload 'apache2.service'.
Authenticating as: Daniel Martínez Regalado (dmartinez)
Password:
==== AUTHENTICATION COMPLETE ====
Job for apache2.service failed.
See "systemctl status apache2.service" and "journalctl -xeu apache2.service" for details.
dmartinez@daniserver:/etc/apache2/sites-available$ sudo a2enmod authz_groupfile
Considering dependency authz_core for authz_groupfile:
Module authz_core already enabled
Enabling module authz_groupfile.
To activate the new configuration, you need to run:
  systemctl restart apache2
dmartinez@daniserver:/etc/apache2/sites-available$ sudo service apache2 restart
dmartinez@daniserver:/etc/apache2/sites-available$ _
```

Denegació d'IP

```
Deny from all
Allow from 172.16.23.0/255.255.255.0
Order deny,allow
```

Cal tenir en compte l'ordre de les IPs